1.1 (1.1) $L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - (ax^2 + by^2)$

$$\begin{cases} m\ddot{x} = -2ax \\ m\ddot{y} = -2by \end{cases} \qquad \begin{cases} P_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} \\ P_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y} \end{cases}$$

$$H = P_x \dot{x} + P_y \dot{y} - L = \frac{1}{2m}(P_x^2 + P_y^2) + (ax^2 + by^2)$$

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial P_x} = \{x, H\} = \frac{1}{m} P_x \\ \dot{y} = \frac{\partial H}{\partial P_y} = \{y, H\} = \frac{1}{m} P_y \end{cases} \qquad \begin{cases} \dot{P}_x = -\frac{\partial H}{\partial x} = \{H, P_x\} = -2ax \\ \dot{P}_y = -\frac{\partial H}{\partial y} = \{H, P_y\} = -2by \end{cases}$$

$E_x = \frac{P_x^2}{2m} + ax^2 \rightarrow \frac{dE_x}{dt} = \cdots = 0$ より) 運動の恒量

↓

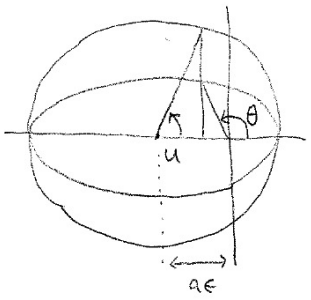
$\frac{x^2}{\frac{E_x}{a}} + \frac{P_x^2}{2mE_x} = 1 \rightarrow \oint P_x dx = \pi \sqrt{\frac{E_x}{a}} \sqrt{2mE_x} = nh + \text{const}$

$\therefore E_x = \frac{nh}{\pi} \sqrt{\frac{a}{2m}} + \text{const}.$

$\frac{P_x}{2m} \dot{P}_x + 2ax\dot{x}$

$\frac{1}{2}\dot{x}\dot{P}_x - \dot{P}_x\dot{x}$

$$p_r = \sqrt{2mE + \frac{2me^2}{r} - \frac{M^2}{r^2}} = \sqrt{-2mE\left(1-\frac{r_1}{r}\right)\left(\frac{r_2}{r}-1\right)}$$



$$\begin{cases} r_1 = a(1-\epsilon) \\ r_2 = a(1+\epsilon) \end{cases} \qquad r_1 + r_2 = 2a = -\frac{e^2}{E}$$

$$r_1 r_2 = a^2(1-\epsilon^2) = -\frac{M^2}{2mE}$$

$$r = \frac{a(1-\epsilon^2)}{1+\epsilon\cos\theta}$$

$$\begin{cases} a = -\dfrac{e^2}{2E} \left(= \dfrac{me^2}{-2mE}\right) \\ \epsilon^2 = 1 + \dfrac{2EM^2}{me^4} < 1 \end{cases}$$

$$r = a(1-\epsilon\cos u)$$

$$\tan\frac{\theta}{2} = \sqrt{\frac{1+\epsilon}{1-\epsilon}}\tan\frac{u}{2}$$

$$\oint p_r\,dr = \sqrt{-2mE}\; a\epsilon^2 \oint \frac{1^2 u}{1-\epsilon\cos u}\,du$$ 有理曲線 (助変数 $t = \tan\frac{u}{2}$)

$$= \sqrt{-2mE}\cdot a\int_0^{2\pi}(1+\epsilon\cos u)\,du - \sqrt{-2mE}\;a(1-\epsilon^2)\int_{u=0}^{u=2\pi}\frac{du}{1-\epsilon\cos u}$$

$$= \frac{me^2}{\sqrt{-2mE}}\cdot(2\pi) - \frac{me^2}{\sqrt{-2mE}}\cdot(1-\epsilon^2)\int_{\theta=0}^{\theta=2\pi}\frac{1}{\sqrt{1-\epsilon^2}}\,d\theta$$

$$= 2\pi\left\{\frac{me^2}{\sqrt{-2mE}} - |M|\right\} \;//$$

☹

$$\sqrt{\frac{1-\epsilon}{1+\epsilon}}\,d\theta = \frac{1+\frac{1-\epsilon}{1+\epsilon}\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\,du$$

$$1-\epsilon\cos u = (1-\epsilon)\frac{1+\tan^2\frac{\theta}{2}}{1+\frac{1-\epsilon}{1+\epsilon}\tan^2\frac{\theta}{2}}$$

1.1 (1.2) $L = \frac{m}{2}(\dot r^2 + r^2\dot\theta^2) + \frac{e^2}{r}$

$$\begin{cases} m\ddot r = -\frac{e^2}{r^2} \\ \frac{d}{dt}(mr^2\dot\theta) = 0 \end{cases} \qquad \begin{cases} P_r = \frac{\partial L}{\partial \dot r} = m\dot r \\ P_\theta = \frac{\partial L}{\partial \dot\theta} = mr^2\dot\theta \end{cases}$$

$$H = P_r\dot r + P_\theta\dot\theta - L = \frac{1}{2m}\left(P_r^2 + \frac{P_\theta^2}{r^2}\right) - \frac{e^2}{r}$$

$$\begin{cases} \dot r = \frac{\partial H}{\partial P_r} = \{r, H\} = \frac{1}{m}P_r \\ \dot\theta = \frac{\partial H}{\partial P_\theta} = \{\theta, H\} = \frac{P_\theta}{mr^2} \end{cases} \qquad \begin{cases} \dot P_r = -\frac{\partial H}{\partial r} = \{H, P_r\} = \frac{2P_\theta^2}{mr^3} - \frac{e^2}{r^2} \\ \dot P_\theta = -\frac{\partial H}{\partial \dot\theta} = \{H, P_\theta\} = 0 \end{cases}$$

H , P_θ

$\oint P_\theta d\theta = 2\pi M$ ($\because P_\theta = \text{const} = M$) $\qquad M = \frac{nh}{2\pi} + \text{const}$

$$\oint P_r dr = 2\int_{r_1}^{r_2}\sqrt{2mE + \frac{2me^2}{r} - \frac{M^2}{r^2}}\,dr = 2\pi\left(\frac{me^2}{\sqrt{-2mE}} - |M|\right) = n'h$$

$$E = -\frac{2\pi^2 me^4}{(|M| + n')^2 h^2}$$

$$\frac{P_r}{m}\dot P_r - \frac{2}{2m}\cdot\frac{P_\theta^2}{r^3}\cdot\dot r + \frac{e^2}{r^2}\dot r$$

$$\dot r\,\dot P_r$$

$$2mE = P_r^2 + \frac{P_\theta^2}{r^2} - \frac{2me^2}{r}$$

$$P_r^2 = 2mE + \frac{2me^2}{r} - \frac{M^2}{r^2}$$

1.2 (1)

$$\int_{-\infty}^{\infty} \forall f(x) \left\{ \lim_{a\to\infty} \frac{a\, g(ax)}{\int_{-\infty}^{\infty} g(x')\,dx'} \right\} dx$$

$$= \lim_{\epsilon\to 0} \frac{\int_{-\infty}^{\infty} f(\epsilon x)\, g(x)\,dx}{\int_{-\infty}^{\infty} g(x)\,dx} = f(0) \qquad (\text{q.e.d})$$

(2) $f(x_i) = 0$

$$\int_{-\infty}^{\infty} \forall g(f(x)) \overbrace{\sum_i \frac{\delta(x-x_i)}{\left|\frac{df(x)}{dx}\right|}}^{\delta(f(x))} df(x) = \sum_i \int_a^b g(f(x))\,\delta(x-x_i)\,dx$$

$[a,b]$ に零点は1つ

$$= \sum_i g(f(x_i)) = \cancel{\sum_i g(0)} = g(0)$$

$\int g(x)\,\delta(f(x))\,dx$ Ⓐ

$$= \int \sum_i \frac{1}{\left|\frac{df(x)}{dx}\right|}\,\delta(x-x_i)\,g(x)\,dx = \sum_i \left|\frac{df(x)}{dx}\right|_{x_i}^{-1} g(x_i)$$ Ⓑ

Ⓐ → Ⓑ を示せばよい

$f(x) = 0$ となる x つまり x_i の近くを考えれば十分

$$Ⓐ = \sum_i \int_{\epsilon_{i-}}^{\epsilon_{i+}} g(x)\,\delta(f(x))\,dx$$

$$g(x)\cdot\delta\left(x_i + \frac{df}{dx}\Delta x\right)$$

1.3 (1) $\hat{A}_1$ は x軸に対する折り返し

$$\hat{A}_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \hat{A}_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\hat{A}_2$ は $y=x$ に対する折り返し

$$\hat{A}_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \hat{A}_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

(2) 自明

(3) $$※ = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix}_{\langle 1, x, x^2 \rangle} \rightarrow \frac{d※}{dx} = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}}_{\sim\sim\sim\sim} ※_{\langle 1, x, x^2 \rangle}$$

(4) 自明 (∵和と実数倍について閉じている)

1.4 (1) $[AB, C] = A[B, C] + [A, C]B$

$[AB, CD] = A[B, CD] + [A, CD]B$

$= A\{C[B, D] + [B, C]D\}$

$+ \{C[A, D] + [A, C]D\}B$

$=$

(2) 与式 $= 0$: Jacobi の恒等式

(3) $[\hat{x}, \hat{p}] = i\hbar$ $\left(\because \frac{d}{dx}x - x\frac{d}{dx} = 1\right)$

$\frac{d}{dx}(xf) = x\frac{df}{dx} + f$

2.2

(1) $$\psi(x) = \int_{-\infty}^{\infty} f(k)\,\psi_k(x)\,dk$$

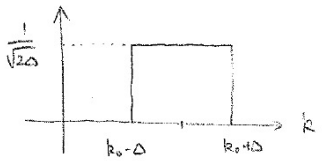
$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(k)\, e^{ikx}\, dk$$

$$f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x)\, e^{-ikx}\, dx$$

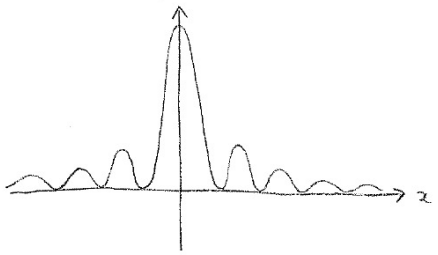
(2) (2.3) $$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(k)\, e^{ikx}\, dk$$

$$= \frac{1}{\sqrt{2\pi}} \int_{k_0-\Delta}^{k_0+\Delta} \frac{1}{\sqrt{2\Delta}}\, e^{ikx}\, dk$$

$$= \sqrt{\frac{\Delta}{\pi}}\, e^{ik_0x}\, \frac{\sin \Delta x}{\Delta x}$$



$$|\psi(x)|^2 = \frac{\Delta}{\pi} \cdot \left(\frac{\sin \Delta x}{\Delta x}\right)^2$$



2.2

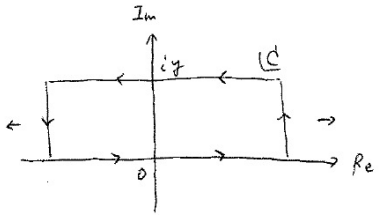
(2) (2.4) $\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(k)\, e^{ikx}\, dk$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{(\pi\Delta^2)^{1/4}} \cdot e^{ikx - \frac{(k-k_0)^2}{2\Delta^2}}\, dk$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(\pi\Delta^2)^{1/4}}\, e^{ik_0 x - \frac{1}{2}\Delta^2 x^2} \int_{-\infty}^{\infty} e^{-\frac{\{k-(k_0+i\Delta^2 x)\}^2}{2\Delta^2}}\, dk$$

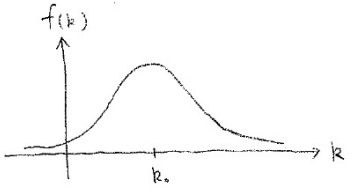
$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$

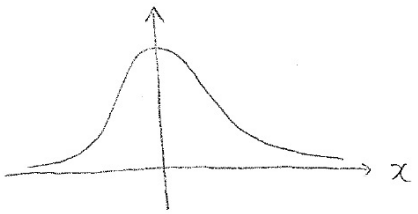
$$= \left(\frac{\Delta^2}{\pi}\right)^{1/4} e^{ik_0 x - \frac{1}{2}\Delta^2 x^2}$$



$$|\psi(x)|^2 = \frac{\Delta}{\sqrt{\pi}}\, e^{-\Delta^2 x^2}$$

$\int_C e^{-\alpha z^2} dz = \sqrt{\frac{\pi}{\alpha}}$



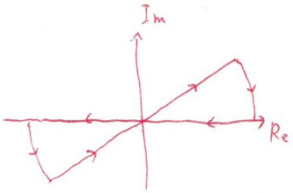


$E = \hbar\omega = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m}$

2.3

(1) $-\frac{(k-k_0)^2}{2\Delta^2} + i\left(kx - \frac{k^2}{2m}\hbar t\right)$

$= -\left(\frac{1}{\Delta^2} + i\frac{\hbar t}{m}\right)\left\{k - \frac{m(\Delta^4 x\hbar t + mk_0) + im\Delta^2(mx - k_0\hbar t)}{\Delta^4\hbar^2t^2 + m^2}\right\}$



$+ \frac{-\Delta^2(mx - k_0\hbar t)^2 + im(\Delta^4x^2\hbar t + 2xk_0m - k_0^2\hbar t)}{2(\Delta^4\hbar^2t^2 + m^2)}$

$= -\frac{1}{2}\left(\frac{1}{\Delta^2} + \frac{i\hbar t}{m}\right)\left(k - \frac{\frac{k_0}{\Delta^2} + ix}{\frac{1}{\Delta^2} + \frac{i\hbar t}{m}}\right)^2 - \frac{k_0^2}{2\Delta^2} + \frac{1}{2}\cdot\frac{\left(\frac{k_0}{\Delta^2} + ix\right)^2}{\frac{1}{\Delta^2} + \frac{i\hbar t}{m}}$

$\Psi(x,t) = \frac{1}{\pi^{1/4}}\cdot\left(\frac{\Delta}{1 + \frac{i\hbar\Delta^2 t}{m}}\right)^{1/2} e^{i\left(k_0x - \frac{\hbar k_0^2}{2m}t\right)} e^{-\frac{\Delta^2}{2}\cdot\frac{\left(x - \frac{k_0\hbar t}{m}\right)^2}{\left(1 + \frac{i\hbar t}{m}\Delta^2\right)}}$

$\langle\hat{x}\rangle = \frac{\hbar k_0}{m}t$

$\Delta x = \frac{1}{\sqrt{2}\Delta}\sqrt{1 + \left(\frac{\hbar\Delta^2}{m}t\right)^2}$

$\langle\hat{p}\rangle = \hbar k_0$ (t によらない → 平面波)

$\Delta p = \frac{\hbar\Delta}{\sqrt{2}}$

(2) $\omega = ck$

$\langle\hat{x}\rangle = ct$

$\Psi(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-a(k-b)^2 + c}\,dk = \frac{1}{\sqrt{2a}}e^c$

2.4

(1) $0 \le \langle\psi|\psi\rangle = \langle\psi|(\hat{x} - i\lambda\hat{p})(\hat{x} + i\lambda\hat{p})|\psi\rangle = \langle\psi|\hat{x}^2 + \lambda^2\hat{p}^2 + i\lambda[\hat{x},\hat{p}]|\psi\rangle$

$= \lambda^2\langle\hat{p}^2\rangle - \hbar\lambda + \langle\hat{x}^2\rangle$

$D \le 0 \rightarrow \langle\hat{x}^2\rangle\langle\hat{p}^2\rangle \ge \frac{\hbar^2}{4}$

(2) $\left.\begin{aligned}\hat{x}' &= \hat{x} - \langle\hat{x}\rangle \\ \hat{p}' &= \hat{p} - \langle\hat{p}\rangle\end{aligned}\right\}$ とおくと $[\hat{x}', \hat{p}'] = [\hat{x}, \hat{p}] = i\hbar$ より (1) から

$\langle\hat{x}'^2\rangle\langle\hat{p}'^2\rangle = (\Delta x)^2(\Delta p)^2 \ge \frac{\hbar^2}{4}$

(3) $|\varphi\rangle = (\hat{A} + i\lambda\hat{B})|\psi\rangle$

$0 \le \langle\hat{A}^2\rangle + \lambda^2\langle\hat{B}^2\rangle + i\lambda[\hat{A},\hat{B}]$

$\rightarrow \langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \ge \frac{1}{4}|[\hat{A},\hat{B}]|^2$

(4) $|(\hat{x}' + i\lambda\hat{p}')|\psi\rangle|^2 = \Delta p^2\left(\lambda - \frac{\hbar}{2\Delta p^2}\right)^2 - \frac{\hbar^2}{4\Delta p^2} + \Delta x^2$

$(\hat{x} + i(2\Delta x^2)\hat{p}')|\psi\rangle = 0$

$-\frac{i}{\hbar}\langle p\rangle\psi + \frac{x}{2\Delta x^2}\psi + \frac{d\psi}{dx} = 0$

$\Delta p \cdot \Delta x = \frac{\hbar}{2}$

$\lambda = \frac{\hbar}{2}\cdot\frac{1}{(\Delta p)^2} = \frac{\Delta x}{\Delta p}$

$\psi = \frac{e^{i\theta}}{(2\pi)^{1/4}(\Delta x)^{1/2}}\, e^{\frac{i}{\hbar}(x - \langle x\rangle)\langle p\rangle} \cdot e^{-\frac{(x - \langle x\rangle)^2}{4\langle x\rangle^2}}$

2.5 $\Psi_n(x,t) = \varphi_n(x) e^{-\frac{i}{\hbar}E_n t}$ $\qquad \Psi = c_n \Psi_n$

$\hat{H}\varphi_n(x) = E_n \varphi_n(x)$ $\qquad \sum_n \varphi_n(x)\varphi_n^*(x) = \delta_{nn'}$

(1) $G(x',t';x,t) = \sum_n \varphi_n(x')\varphi_n^*(x) e^{\frac{i}{\hbar}E_n(t-t')} = \sum_n \langle x'|n\rangle\langle n|x\rangle e^{-\frac{i}{\hbar}E_n(t'-t)}$

$= \langle x'| e^{-\frac{i}{\hbar}\hat{H}(t'-t)} |x\rangle = \int dk \langle x'| e^{-\frac{i}{\hbar}\hat{H}(t'-t)} |k\rangle\langle k|x\rangle$ ($\hat{H} \to \frac{\hbar^2k^2}{2m}$)

$= \sum_n \Psi_n(x',t')\Psi_n^*(x,t)$

$= \int_{-\infty}^{\infty} dk\, e^{-\frac{i}{\hbar}\frac{\hbar^2k^2}{2m}(t'-t)} \langle x'|k\rangle\langle k|x\rangle = \sqrt{\frac{m}{2\pi i\hbar(t'-t)}}\, e^{\frac{i}{\hbar}\cdot\frac{m}{2}\cdot\frac{(x'-x)^2}{t'-t}}$ ($\langle x'|k\rangle = \frac{1}{\sqrt{2\pi}}e^{ikx'}$, $\langle k|x\rangle = \frac{1}{\sqrt{2\pi}}e^{-ikx}$)

$\to \delta(x-x')$ $\quad (t' \to t)$

$\to \frac{1}{2\pi}\int_{-\infty}^{\infty} dk\, e^{ik(x-x')}$

(2) $\int G(x',t';x,t)\Psi(x,t)\,dx = \sum_n c_n \Psi_n(x',t') \int \Psi_n^*(x,t)\Psi_n(x,t)\,dx$

$\int dx\, G\langle x|\alpha\rangle = \int d\xi \int dx \langle x'| e^{-\frac{i}{\hbar}\hat{H}(t'-t)} |\xi\rangle\langle\xi|x\rangle\langle x|\alpha\rangle = \int d\xi\, e^{-\frac{i}{\hbar}E_\xi(t'-t)}\langle x'|\xi\rangle\langle\xi|\alpha\rangle$ ($\Psi(x,t)$, $= \Psi(x',t')$)

(3) $\int dx \langle x'| e^{-\frac{i}{\hbar}\hat{H}t'} |x\rangle\langle x|\alpha\rangle = \langle x'| e^{-\frac{i}{\hbar}\hat{H}t'} |\alpha\rangle = \int dk \langle x'| e^{-\frac{i}{\hbar}\hat{H}t'} |k\rangle\langle k|\alpha\rangle$ ($\langle x'|\alpha\rangle = \Psi(x,0)$)

$\Psi(x,0) = \langle x|\alpha\rangle = \int dk \langle x|k\rangle\langle k|\alpha\rangle = \int dk \frac{1}{\sqrt{2\pi}} e^{ikx}\langle k|\alpha\rangle$

$\langle k|\alpha\rangle = (\pi\Delta^2)^{-1/4} e^{-\frac{(k-k_0)^2}{2\Delta^2}}$

$|\Psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\Psi(0)\rangle$

2.6

(1) $\hat{A}\psi = a\psi$

$\langle \hat{A}\psi | \psi \rangle = \langle \psi | \hat{A}^{+}\psi \rangle$

$a^{*}\langle \psi | \psi \rangle = a\langle \psi | \psi \rangle$

(2) $\int (\hat{P}\psi)^{*}\phi \, dx = \int \left(\frac{\hbar}{i}\frac{\partial \psi}{\partial x}\right)^{*}\phi \, dx$

$= \int -\frac{\hbar}{i}\frac{\partial \psi^{*}}{\partial x}\phi \, dx$

$= \left[-\frac{\hbar}{i}\psi^{*}\phi\right] + \int \psi^{*}\frac{\hbar}{i}\frac{\partial \phi}{\partial x} dx$

$= \int \psi^{*}\hat{P}\phi \, dx$

$\hat{P}$ は エルミート

(3) $\frac{d}{dt}\int \psi^{*}\psi \, dx = \frac{1}{i\hbar}\int (\psi^{*}H\psi - \psi H\psi^{*}) \, dx = 0$

3.1

(1) $\hat{p}\,u_p = \varepsilon u_p \qquad u_p = e^{\frac{i}{\hbar}\varepsilon x}$

$\hat{x}\,u_x = \varepsilon u_x \qquad u_x = \delta(x-\varepsilon)$

(2) $\psi(x) = \int f(p)\,u_p(x)\,dp \qquad f(p) = \int \psi(x)\,u_p^*(x)\,dx$

$= \int \delta(x-x')\,\frac{1}{\sqrt{2\pi\hbar}}\,e^{\frac{i}{\hbar}p(x-x')}\,dx = \frac{1}{\sqrt{2\pi\hbar}}\,e^{\frac{i}{\hbar}px'}$

(3) $\int f^*(p)\,p\,g(p)\,dp = \int dp\,dx\,dx'\,(\psi(x')u_p^*(x'))^*\,p\,(\varphi(x)u_p^*(x))$

$= \int dp\,dx\,dx'\,u_p(x')\,\psi^*(x')\,p\,\varphi(x)\,u_p^*(x)$

$= \int dx\,dx'\,\psi^*(x')\,\varphi(x)\,\frac{1}{2\pi\hbar}\int dp\,p\,e^{\frac{i}{\hbar}p(x-x')}$

$= \frac{1}{2\pi\hbar}\int dx\,dx'\,\psi^*(x')$

(4) $\int \psi^*(x)\,F(\hat{x})\,\varphi(x)$

$(\hat{x})^n e^{\frac{i}{\hbar}p'x} = x^n e^{\frac{i}{\hbar}p'x} = \left(i\hbar\frac{\partial}{\partial p'}\right)^n e^{\frac{i}{\hbar}p'x} \qquad (\hat{x})^n = \left(i\hbar\frac{\partial}{\partial p'}\right)^n$

3.2

(1) $\frac{dW}{dx} = \psi_1\psi_2'' - \psi_1''\psi_2 = (V(x) - \epsilon)(\psi_1\psi_2 - \psi_1\psi_2) = 0$

(2) $W =$ const with x $\quad \psi \to 0$ $(|x| \to \infty)$ 时 $W = 0$ $\quad \therefore \frac{\psi_1}{\psi_2} =$ const with x

(3) $W' = (\epsilon_1 - \epsilon_2)\psi_1\psi_2$

3.3 $\psi''(x) + (\epsilon - V(x))\psi(x) = 0$ — $V \sim O(|x|^{-1-\epsilon})$

$W = \psi_1\psi_2' - \psi_1'\psi_2$ $\qquad k(x) = \sqrt{\epsilon - V(x)}$

	$x \to -\infty$	$x \to +\infty$
ψ_1 :	$Ae^{i(kx-\omega t)} + Be^{i(-kx-\omega t)}$	$Ce^{i(kx-\omega t)}$
ψ_2 :	$C'e^{i(-kx-\omega t)}$	$A'e^{i(-kx-\omega t)} + B'e^{i(kx-\omega t)}$

$W(-\infty) = W(+\infty) \to AC' = A'C \quad \left(\frac{C'}{A} = \frac{C}{A}\right)$

~~$T'(-ik)$~~

$\left(e^{ikx} + Re^{-ikx}\right)(-ik)T'e^{-ikx} - ik\left(e^{ikx} - Re^{-ikx}\right)T'e^{-ikx}$

	$-\infty$	$+\infty$
ψ_1	$\frac{1}{\alpha} + R\alpha$	$T\alpha$
ψ_1'	$ik\left(\frac{1}{\alpha} - R\alpha\right)$	$ikT\alpha$
ψ_2	$T'\alpha$	$R'\alpha + \frac{1}{\alpha}$
ψ_2'	$-ikT'\alpha$	$ik\left(R'\alpha - \frac{1}{\alpha}\right)$

3.4. $\int \psi^* \psi \, dx < \infty$

(1) $E > 0$ → $\int \psi^* \psi \, dx \geq \int_{x_2}^{\infty} |\psi|^2 dx = \infty$

$E = 0$ → $[x_1, x_2]$ で $\int \psi^* \psi dx < \infty$ → $\psi = 0$

$\psi(x_1) = \psi'(x_1) = 0$ $\quad \psi'' = \frac{2m}{\hbar^2} V \psi$ より ψ と ψ'' は異符号 → $\psi''(x_1) = 0$

$\therefore \psi''(x) = 0$

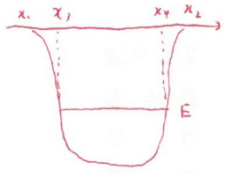
$E < 0$ → $\quad x < x_1 : \psi(x) = A \exp\left(\frac{\sqrt{-2mE}}{\hbar} x\right)$

$(E \leq -V_0)$ $\quad x_2 < x : \psi(x) = B \exp\left(-\frac{\sqrt{-2mE}}{\hbar} x\right)$

…なだらかに 0 に収束 (ψ と ψ'' は同符号)

$\therefore -V_0 < E < 0$

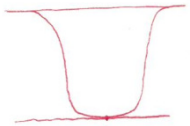
(2)



$\psi_E(x) \to \psi(x+\Delta x) + \frac{\partial \psi_E(x+\Delta x)}{\partial E} \Delta E$ $\quad \frac{\Delta x}{\Delta E}$ ↓

$= \psi_E(x) + \psi_E'(x) \Delta x + \left(\frac{\partial \psi_E(x)}{\partial E} + \frac{\partial^2 \psi_E(x)}{\partial E \partial x} \frac{\partial V}{\partial x} \Delta E \right) \Delta E$

(4)



↑
ここで粒子の位置が決まる

$E = \langle \psi | H | \psi \rangle$

$= \frac{1}{2m} \langle \psi | p^2 | \psi \rangle + \langle \psi | V | \psi \rangle$

$= \frac{1}{2m} \underbrace{\langle \psi | p^2 | \psi \rangle}_{\geq 0} - V_0 \quad > -V_0$

$(\Delta p)^2$ $\qquad \Delta p \Delta x \geq \frac{\hbar}{2}$

3.5

(1) $\psi'' + (E - V(x))\psi = 0$

$k_- = \sqrt{E+V_0}$ $\quad \psi_- = A_- e^{ik_- x}$ $\quad \tilde{\psi}_- = \tilde{A} e^{-ik_- x}$

$k_+ = \sqrt{E}$ $\quad \psi_+ = A_+ e^{ik_+ x}$

Boundary Condition $(x=0)$

$\psi_- + \tilde{\psi}_- = \psi_+ \rightarrow A_- + \tilde{A} = A_+$

$\psi_-' + \tilde{\psi}_-' = \psi_+' \rightarrow k_- A_- - k_- \tilde{A} = k_+ A_+$

$R = \left|\frac{\tilde{A}}{A_-}\right|^2 = \left(\frac{k_- - k_+}{k_- + k_+}\right)^2 = \left(\frac{1-\sqrt{1-V_0/E}}{1+\sqrt{1-V_0/E}}\right)^2$

$T = \left(\frac{A_+}{A_-}\right)^2 = \left(\frac{2k_-}{k_- + k_+}\right)^2 = \left(\frac{2\sqrt{1-V_0/E}}{1+\sqrt{1-V_0/E}}\right)^2$

(2) 波数が虚数 → 反射

(3) $v_g = \frac{\partial \omega}{\partial k} = \frac{\hbar}{m} k$ $\left(= \frac{p}{m}\right)$ より $\quad T_{eff} = \frac{k_+}{k_-}\left(\frac{A_+}{A_-}\right)^2 = \frac{4k_+k_-}{(k_- + k_-)^2}$

$(E<0)$

$k_+ = ik$

確率の流れ $T = 0$

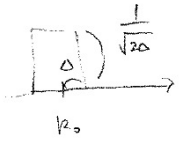
$R = \left|\frac{k_0 - ik}{k_0 + ik}\right|^2 = 1$

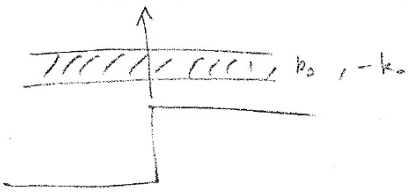
$T = \left|\frac{4k_0 k}{(k_0 + ik)^2} i\right| = \frac{4k_0 k}{k_0^2 + k^2}$

3.6 $\psi(x) \sim \begin{cases} e^{ikx} + C(k)e^{-ikx} & (x<0) \\ D(k)e^{ikx} & (x>0) \end{cases}$

$\varphi(x) = \int f(k)\,\psi_k(x)\,dx$

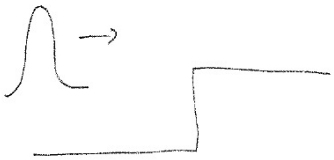
$g(x,t) = \int f(k)\,e^{i(kx-\omega(k)t)}\,dk$





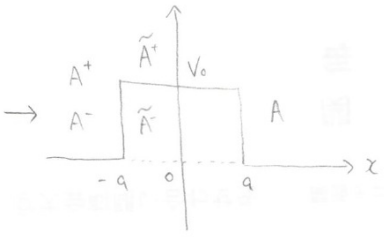
$f(k) = \delta(k-k_0) + \delta(k+k_0)$ X

$\psi = \frac{1}{\sqrt{2\pi}} \int dk\, f(k)\, e^{i\left(kx - \frac{\pm\hbar^2}{2m}t\right)}$



$|\psi,t\rangle = e^{-\frac{i}{\hbar}Ht}\,|\psi\rangle$

3.7



$k = \sqrt{E}$

$\tilde{k} = \sqrt{E - V_0}$

(1)
$$\begin{cases} A^+ e^{ik(-a)} + A^- e^{-ik(-a)} = \tilde{A}^+ e^{i\tilde{k}(-a)} + \tilde{A}^- e^{-i\tilde{k}(-a)} \\ k\left(-A^+ e^{-ika} + A^- e^{ika}\right) = \tilde{k}\left(-\tilde{A}^+ e^{-i\tilde{k}a} + \tilde{A}^- e^{i\tilde{k}a}\right) \end{cases}$$

$$\begin{cases} \tilde{A}^+ e^{i\tilde{k}a} + \tilde{A}^- e^{-i\tilde{k}a} = A e^{ika} \\ \tilde{k}\left(\tilde{A}^+ e^{i\tilde{k}a} - \tilde{A}^- e^{-i\tilde{k}a}\right) = k A e^{ika} \end{cases}$$

$\alpha = e^{ika}$

$$M = \begin{pmatrix} \frac{1}{\alpha} & \alpha \\ -\frac{k}{\alpha} & \alpha k \end{pmatrix}$$

$$\vec{A} = \begin{pmatrix} A^+ \\ A^- \end{pmatrix}$$

$$\left.\begin{array}{l} M\vec{A} = \tilde{M}\vec{\tilde{A}} \\ \tilde{M}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\vec{\tilde{A}} = A\alpha\begin{pmatrix} 1 \\ k \end{pmatrix} \end{array}\right\} \quad \begin{pmatrix} A^+ \\ A^- \end{pmatrix} = M^{-1}\tilde{M}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\tilde{M}^{-1}M\begin{pmatrix} 0 \\ A \end{pmatrix} = \frac{A\alpha}{4\tilde{\alpha}^2 k\tilde{k}}\begin{pmatrix} \alpha^2\left\{\tilde{\alpha}^4(k-\tilde{k})^2 - (k+\tilde{k})^2\right\} \\ -(k^2-\tilde{k}^2)(\tilde{\alpha}^4-1) \end{pmatrix}$$

$$R = \left|\frac{A^-}{A^+}\right|^2 = \left|\frac{-(k^2-\tilde{k}^2)(\tilde{\alpha}^4-1)}{\alpha^2\left\{\tilde{\alpha}^4(k-\tilde{k})^2 - (k+\tilde{k})^2\right\}}\right|^2 = \left|\frac{4k\tilde{k}}{(\tilde{k}+k)^2 e^{2i(\tilde{k}-k)a} - (\tilde{k}-k)^2 e^{2i(\tilde{k}+k)a}}\right|^2$$

$$T = \left|\frac{A}{A^+}\right|^2 = \left|\frac{4\tilde{\alpha}^2 k\tilde{k}}{\alpha^2\left\{\tilde{\alpha}^4(k-\tilde{k})^2 - (k+\tilde{k})^2\right\}}\right|^2 = \left|\frac{2i(k^2-\tilde{k}^2)\sinh 2\tilde{k}a}{\ldots}\right|^2$$

$$A^+ e^{ika} + A^- e^{-ika} = \tilde{A}^+\cosh\tilde{k}(-a) + \tilde{A}^-\sinh\tilde{k}(-a)$$

$$\tilde{A}^+\cosh\tilde{k}a + \tilde{A}^-\sinh\tilde{k}a = A e^{ika}$$

$$T = \left|\frac{e^{-2ika}}{\cosh 2\tilde{k}a + \frac{i}{2}\left(\frac{\tilde{k}}{k} - \frac{k}{\tilde{k}}\right)\sinh 2\tilde{k}a}\right|^2$$

$$R = \left|\frac{-\frac{i}{2}\left(\frac{\tilde{k}}{k} + \frac{k}{\tilde{k}}\right)\sinh 2ka}{\ldots}\right|^2$$

3.7

(2) $R = 0 \longrightarrow \sin^2 2\tilde{k}a = 0$

$4\tilde{k}a = 2\pi n \qquad E = V_0 + \dfrac{n^2\pi^2\hbar^2}{8ma^2} \qquad \lambda = \dfrac{4a}{n}$

(3) $E \gg V_0$ のとき

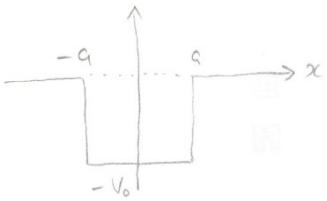
$$\psi(x) = \frac{4k\tilde{k}}{(k+\tilde{k})^2 e^{2i(k-\tilde{k})a} - (k-\tilde{k})^2 e^{2i(k+\tilde{k})a}} e^{ikx} \propto e^{ikx}\left\{(k+\tilde{k})e^{-2i(k-\tilde{k})a} - (k-\tilde{k})^2 e^{-2i(k+\tilde{k})a}\right\}$$

$$\tilde{k} = \sqrt{\frac{2m}{\hbar^2}(E-V_0)} \fallingdotseq k - \underbrace{\sqrt{\frac{2mE}{\hbar^2}}\frac{1}{2}\frac{V_0}{E}}_{\Delta k} \qquad \psi(x) \propto e^{i(kx+2\Delta k a)}$$

$$\delta = -2a\Delta k = \frac{-V_0}{E}\sqrt{\frac{2mE}{\hbar^2}}\,a$$

3.8

(1)



$\hbar^2 k^2 = 2m(E+V_0)$ $\quad \hbar^2\lambda^2 = -2mE$

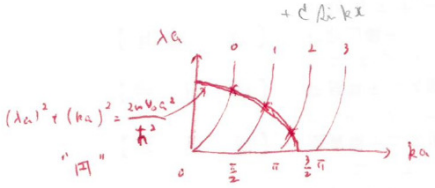
$Ae^{\lambda x}$ $\quad B\cos kx + C\sin kx$ $\quad De^{-\lambda x}$

even: $C=0$, $A=D$ $\quad$ odd: $B=0$ $\quad A=-D$

$B\cos ka = De^{-\lambda a}$ $\quad C\sin ka = De^{-\lambda a}$

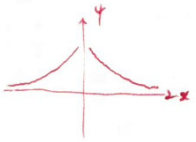
$-kB\sin ka = -\lambda De^{-\lambda a}$ $\quad kC\cos ka = -\lambda De^{-\lambda a}$

$ka\tan(ka) = \lambda a$ $\quad ka\cot(ka) = -\lambda a$



(2) $ka \to 0$ 近似 $\quad \lambda a = ka\tan(ka)$

$\doteqdot k^2a^2$ $\quad \rightarrow \quad E = -\frac{ma^2}{2\hbar^2}$



(3) $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - a\delta(x)\psi = E\psi$

$-\frac{\hbar^2}{2m}\int_{-\epsilon}^{\epsilon}\frac{d^2\psi}{dx^2}dx - a\int_{-\epsilon}^{\epsilon}\delta(x)\psi(x)dx = E\int_{-\epsilon}^{\epsilon}\psi(x)dx$

$\psi(0)$ $\quad$ 0

$\psi = \begin{cases} \psi(0)e^{-\lambda x} & (x>0) \\ \psi(0)e^{\lambda x} & (x<0) \end{cases}$

$\frac{\hbar^2}{2m}\left\{\left.\frac{d\psi}{dx}\right|_{x=+0} - \left.\frac{d\psi}{dx}\right|_{x=-0}\right\} + a\psi(0) = 0$

$\frac{\hbar^2}{2m}\cdot 2\lambda + a = 0$

$E = -\frac{ma^2}{2\hbar^2}$

4.2

(1) $H\psi_n = E_n\psi_n \qquad H(\hat{a}^+\psi_n) = (E_n + \hbar\omega)(\hat{a}^+\psi_n) \qquad$ if $\psi_n = \frac{1}{\sqrt{n!}}(\hat{a}^+)^n\psi_0 \qquad E_n = (n+\frac{1}{2})\hbar\omega$

$$\langle\psi_n|\psi_n\rangle = \frac{1}{n!}\langle\psi_0|(\hat{a})^n(\hat{a}^+)^n|\psi_0\rangle$$

$$[\hat{a},(\hat{a}^+)^n] = n(\hat{a}^+)^{n-1}$$

$$= \frac{1}{(n-1)!}\langle\psi_0|(\hat{a})^{n-1}(\hat{a}^+)^{n-1}|\psi_0\rangle + \frac{1}{n!}\langle\psi_0|(\hat{a})^{n-1}(\hat{a}^+)^n\,\underbrace{\hat{a}|\psi_0\rangle}_{0}$$

$$= \dots = \langle\psi_0|\psi_0\rangle = 1$$

(2) $$e^{\hat{S}}\hat{O}e^{-\hat{S}} = \left(1+\hat{S}+\frac{1}{2}(\hat{S})^2+\dots+\frac{1}{n!}(\hat{S})^n+\dots\right)\hat{O}\left(1-\hat{S}+\frac{1}{2}(\hat{S})^2-\dots+\frac{(-1)^n}{n!}(\hat{S})^n+\dots\right)$$

$$\hat{F}(t) = e^{t\hat{S}}\hat{O}e^{-t\hat{S}} \qquad \frac{d\hat{F}(t)}{dt} = e^{\hat{S}t}[\hat{S},\hat{O}]e^{-\hat{S}t}$$

$$\frac{d^n\hat{F}(t)}{dt^n} = e^{\hat{S}t}[\underbrace{\hat{S}[\hat{S}\cdots[\hat{S}}_{n},\hat{O}]]\}\}]e^{-\hat{S}t}$$

(3) $$\left[\tfrac{1}{2}a^2\hat{x}^2,\hat{p}\right] = \frac{a^2}{2}[\hat{x}^2,\hat{p}] = a^2\hat{x}\,i\hbar$$

$$-\frac{i}{\sqrt{2m\hbar\omega}}e^{\frac{a^2\hat{x}^2}{2}}\hat{p}\,e^{-\frac{a^2\hat{x}^2}{2}} = -\frac{i}{\sqrt{2m\hbar\omega}}(\hat{p}+i\hbar a^2\hat{x}+\dots) = \hat{Q}-i\hat{P} = \hat{a}^+ \qquad (\text{rest} = 0)$$

$$\hat{p} = -i\hbar\frac{d}{dx}$$

$$\psi_n(x) = \frac{1}{\sqrt{n!}}(\hat{a}^+)^n\psi_0(x) = \frac{1}{\sqrt{n!}}\left(\frac{-i}{\sqrt{2m\hbar\omega}}\right)^n\left(e^{\frac{a^2\hat{x}^2}{2}}\hat{p}\,e^{-\frac{a^2\hat{x}^2}{2}}\right)^n\psi_0(x) = \frac{1}{\sqrt{n!}}\left(\sqrt{\frac{\hbar}{2m\omega}}\right)^n\left[e^{\frac{a^2\hat{x}^2}{2}}\frac{d^n}{d\hat{x}^n}e^{-\frac{a^2\hat{x}^2}{2}}\right]\psi_0(x)$$

$$= \frac{1}{\sqrt{n!}}\cdot\frac{1}{(\sqrt{2a})^n}$$

(4) $a|n\rangle = \sqrt{n}\,|n-1\rangle \qquad a^+|n\rangle = \sqrt{n+1}\,|n+1\rangle \qquad \langle m|n\rangle = \delta_{mn}$

$$\langle m|a|n\rangle = \sqrt{n}\,\delta_{m,n-1} \qquad \langle m|a^+|n\rangle = \sqrt{n+1}\,\delta_{m,n+1}$$

$$\langle m|\hat{Q}|n\rangle = Q_{mn} = \left(\frac{\sqrt{n}}{2}\delta_{m,n-1}+\frac{\sqrt{n+1}}{2}\delta_{m,n+1}\right)$$

$$\langle m|\hat{P}|n\rangle = P_{mn} = \frac{1}{2i}\left(\sqrt{n}\,\delta_{m,n-1}-\sqrt{n+1}\,\delta_{m,n+1}\right)$$

$$x_{mn} = \sqrt{\frac{2\hbar}{m\omega}}Q_{mn} \qquad P_{mn} = \sqrt{2m\hbar\omega}\,P_{mn}$$

$$\langle n|\frac{p^2}{2m}|n\rangle = \hbar\omega\langle n|\hat{P}^2|n\rangle = \hbar\omega\left(\frac{n}{2}+\frac{1}{4}\right) \qquad \left(\frac{\hbar\omega}{4}\langle n|(-\hat{a}^2-\hat{a}^{+2}+\hat{a}\hat{a}^++\hat{a}^+\hat{a})|n\rangle\right)$$

$$\langle n|\hbar\omega\hat{Q}^2|n\rangle = \frac{1}{2}\langle n|\hat{H}|n\rangle = \hbar\omega\left(\frac{1}{2}+\frac{1}{4}\right)$$

$$\Delta Q\Delta P = \frac{n}{2}+\frac{1}{4}$$

$$\Delta p\Delta x = 2\hbar\Delta P\Delta Q = \hbar\left(n+\frac{1}{2}\right)$$

5.1

$$[\hat{L}_\pm, \hat{L}_z] = [\hat{L}_x, \hat{L}_z] \pm i[\hat{L}_y, \hat{L}_z] = -i\hat{L}_y \mp \hat{L}_x = \mp \hat{L}_\pm \;//$$

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z]$$

$$= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y$$

$$= -i\hat{L}_x\hat{L}_y - i\hat{L}_y\hat{L}_x + i\hat{L}_y\hat{L}_x + i\hat{L}_x\hat{L}_y = 0 \;//$$

$$[\hat{L}^2, \hat{L}_\pm] = [\hat{L}^2, \hat{L}_x] \pm i[\hat{L}^2, \hat{L}_y] = 0 \;//$$

(2) $r^2 = x^2 + y^2 + z^2$ $\quad \tan\theta = \dfrac{\sqrt{x^2+y^2}}{z}$ $\quad \tan\phi = \dfrac{y}{x}$

$$L_x = \frac{1}{i}\left\{-\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right\}$$

$$L_y = \frac{1}{i}\left\{\cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right\}$$

$$L_z = \frac{1}{i}\frac{\partial}{\partial\phi}$$

$$L_\pm = e^{\pm i\phi}\left\{\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right\}$$

$$\mathbb{L} = -\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right\}$$

5.1

$[x, p] = i\hbar$

(1) $\hat{L}_k = \frac{1}{\hbar}(\hat{r}\times\hat{p})_k = \frac{1}{\hbar}\hat{r}_i\hat{p}_j\epsilon_{ijk}$

$$[\hat{L}_k, \hat{r}_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{\hat{r}_i\underbrace{[\hat{p}_j, \hat{r}_\ell]}_{-i\hbar\delta_{j\ell}} + \underbrace{[\hat{r}_i, \hat{r}_\ell]}_{0}\hat{p}_j\right\} = -i\hat{r}_i\epsilon_{i\ell k}$$

$$[\hat{L}_k, \hat{p}_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{\hat{r}_i\underbrace{[\hat{p}_j, \hat{p}_\ell]}_{0} + \underbrace{[\hat{r}_i, \hat{p}_\ell]}_{i\hbar\delta_{i\ell}}\hat{p}_j\right\} = i\hat{p}_j\epsilon_{\ell jk}$$

$$[\hat{L}_k, \hat{L}_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{\hat{r}_i\underbrace{[\hat{p}_i, \hat{L}_\ell]}_{-i\hat{p}_m\epsilon_{jm\ell}} + \underbrace{[\hat{r}_i, \hat{L}_\ell]}_{i\hat{r}_n\epsilon_{ni\ell}}\hat{p}_j\right\}$$

$$= -\frac{i}{\hbar}\left(\hat{r}_i\hat{p}_m\underbrace{\epsilon_{ijk}\epsilon_{jm\ell}}_{\epsilon_{kij}\epsilon_{m\ell j}} - \hat{r}_n\hat{p}_j\underbrace{\epsilon_{ijk}\epsilon_{ni\ell}}_{\epsilon_{jki}\epsilon_{\ell ni}}\right)$$

$$= -\frac{i}{\hbar}\hat{r}_i\hat{p}_m(\delta_{km}\delta_{i\ell} - \delta_{k\ell}\delta_{im}) + \frac{i}{\hbar}\hat{r}_n\hat{p}_j(\delta_{j\ell}\delta_{kn} - \delta_{jn}\delta_{k\ell})$$

$=$

5.1

$$L_k = \frac{1}{\hbar}(r \times p)_k = \frac{1}{\hbar} r_i p_j \epsilon_{ijk}$$

$$i \epsilon_{klm} L_m$$

$$i \epsilon_{klm} \frac{1}{\hbar} r_s p_t \epsilon_{stm}$$

$$[L_k, r_l] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, r_l]}_{-i\hbar\delta_{jl}} + \underbrace{[r_i, r_l]}_{0} p_j \right\} = -i r_i \epsilon_{isk}$$

$$[L_k, p_l] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, p_l]}_{0} + \underbrace{[r_i, p_l]}_{i\hbar\delta_{il}} p_j \right\} = i p_j \epsilon_{ljk}$$

$$[L_k, L_l] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, L_s]}_{-i p_m \epsilon_{jml}} + \underbrace{[r_i, L_s]}_{i r_n \epsilon_{nil}} p_j \right\} = \frac{i}{\hbar}\left(- r_i p_m \underbrace{\epsilon_{ijk}\epsilon_{jml}}_{\epsilon_{kij}\epsilon_{mlj}} + r_n p_j \underbrace{\epsilon_{ijk}\epsilon_{nil}}_{\epsilon_{jki}\epsilon_{lni}} \right)$$

$\frac{\epsilon_{ijk}}{\hbar} r_i p_j$

$$= -\frac{i}{\hbar} r_i p_m (\delta_{km}\delta_{il} - \delta_{kl}\delta_{im}) + \frac{i}{\hbar} r_n p_j (\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl})$$

$\alpha_i \delta_{ij} = \alpha_j$

$$= -\frac{i}{\hbar} r_l p_k + \underbrace{\frac{i}{\hbar} r_i p_i \delta_{kl}} + \frac{i}{\hbar} r_k p_l \underbrace{- \frac{i}{\hbar} r_j p_j \delta_{kl}}$$

$$= \frac{i}{\hbar}(r_k p_l - r_l p_k)$$

5.2

(1) $\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$, $\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + \left(-\frac{m^2}{\sin^2\theta} + l(l+1)\right)\Theta = 0$ $L^2 = L_x^2 + L_y^2 + L_z^2$

$L^2 - L_z^2 = L_x^2 + L_y^2$

(2) $[L^2, L_\pm] = 0$ $L^2(L_\pm Y_{lm}) = L_\pm L^2 Y_{lm} = l(l+1)(L_\pm Y_{lm})$ $l(l+1) - m^2 \geq 0$

$[L_\pm, L_z] = \mp L_\pm$ $L_z(L_\pm Y_{lm}) = (L_\pm L_z \pm L_\pm) Y_{lm} = (m \pm 1)(L_\pm Y_{lm})$

(3) $L_+ Y_{lm} = 0$ $\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)\Theta e^{im_{max}\phi} = 0$ → $\frac{d\Theta}{d\theta} = m_{max}\cot\theta\,\Theta$ → $\Theta = c(\sin\theta)^{m_{max}}$

$e^{im\phi}\frac{d\Theta}{d\theta} + i\cot\theta\cdot\Theta\, im e^{im\phi} = 0$

$\{m(m+1) - l(l+1)\}\sin^m\theta = 0$ $m_{max} = l$ $\frac{d\Theta}{\Theta} = m\cot\theta\, d\theta = m\frac{\cos\theta}{\sin\theta}d\theta$

$\ln\Theta =$

(4) $Y_{ll} = c_l e^{il\phi}\sin^l\theta$

$Y_{l,l-1} = c_l e^{i\phi}\left(-\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)e^{il\phi}\sin^l\theta = c_l e^{i(l-1)\phi}($

$c_l e^{i\phi}\left(-l\cos\theta\sin^{l-1}\theta\, e^{il\phi} + i\frac{\cos\theta}{\sin\theta}(il)e^{il\phi}\sin^l\theta\right)$

$= c_l\left(-l\cos\theta\sin^{l-1}\theta - l\cos\theta\sin^{l-1}\theta\right)e^{i(l-1)\phi}$

$Y_{lm} = (-1)^l\sqrt{\frac{(l+m)!\,(2l+1)!!}{4\pi\, 2^l l!\,(l-m)!\,(2l)!}}\, e^{im\phi}\sin^m\theta\frac{d^{l-m}}{d(\cos\theta)^{l-m}}\sin^{2l}\theta$

(5) $l = 1$, $m = -1, 0, 1$

$Y_{00} =$

5.3

$J_+ J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2$

(1) $J_\pm |jm\rangle$ の ノルム

$\langle jm| J_\mp J_\pm |jm\rangle = \{\lambda_j - m(m\pm1)\}\langle jm|jm\rangle \geq 0$ $\quad \lambda_j \geq m(m\pm1)$

$J_\pm |\lambda_j, \overset{m_{max}}{m_{min}}\rangle = 0$ $\quad \frac{-1-\sqrt{1+4\lambda}}{2} \leq m \leq \frac{-1+\sqrt{1+4\lambda}}{2}$

(2) $J_+ |\lambda_j\, m_{max}\rangle = 0$

$J_- J_+ |\lambda_j\, m_{max}\rangle = (J^2 - J_z(J_z+1))|\lambda_j, m_{max}\rangle$

$= (\lambda_j - m_{max}(m_{max}+1))|\lambda_j, m_{max}\rangle = 0$

$J_- |\lambda_j\, m_{min}\rangle = 0$ $\quad \lambda_j = m_{max}(m_{max}+1)$

$J_+ J_- |\lambda_j, m_{min}\rangle = (\lambda_j - m_{min}(m_{min}-1))|\lambda_j, m_{min}\rangle = 0$

$m_{max}(m_{max}+1) = m_{min}(m_{min}-1) \rightarrow (m_{max} + m_{min})(m_{max} - m_{min} + 1) = 0$

$\overset{\vee}{0}$ $\qquad\qquad$ $\overset{\vee}{0}$

(3) $J_\pm |jm\rangle = \sqrt{(j\mp m)(j\pm m+1)}\, |j, m\pm1\rangle$

$\langle j'm'| J_\pm |jm\rangle = \sqrt{(j\pm m)(j\pm m+1)}\, \delta_{jj'}\delta_{m',m\pm1}$

$\langle j'm'| J_z |jm\rangle = m\,\delta_{jj'}\delta_{mm'}$

$j=\frac{1}{2}$ $\quad J_z = \begin{pmatrix} \frac{1}{2} & \\ & -\frac{1}{2} \end{pmatrix}$, $J_+ = \begin{pmatrix} & 1 \\ & \end{pmatrix}$ $\quad J_- = \begin{pmatrix} & \\ 1 & \end{pmatrix}$

$J_x = \begin{pmatrix} & \frac{1}{2} \\ \frac{1}{2} & \end{pmatrix}$ $\quad J_y = \begin{pmatrix} & -\frac{i}{2} \\ \frac{i}{2} & \end{pmatrix}$

$j=1$ $\quad J_z = \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}$ $\quad J_+ = \begin{pmatrix} & \sqrt{2} & \\ & & \sqrt{2} \\ & & \end{pmatrix}$ $\quad J_- = \begin{pmatrix} & & \\ \sqrt{2} & & \\ & \sqrt{2} & \end{pmatrix}$

$J_x = \begin{pmatrix} & \frac{1}{\sqrt{2}} & \\ \frac{1}{\sqrt{2}} & & \frac{1}{\sqrt{2}} \\ & \frac{1}{\sqrt{2}} & \end{pmatrix}$ $\quad J_y = \begin{pmatrix} & -\frac{i}{\sqrt{2}} & \\ \frac{i}{\sqrt{2}} & & -\frac{i}{\sqrt{2}} \\ & \frac{i}{\sqrt{2}} & \end{pmatrix}$

(4) $\sigma_x = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$ $\quad \sigma_y = \begin{pmatrix} & -i \\ i & \end{pmatrix}$ $\quad \sigma_z = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$

5.4

(1) $\Delta = \frac{1}{r^2}\partial_r\left(r^2\partial_r\right) - \frac{\hat{L}^2}{r^2}$

$\partial_r(r^2\partial_r\psi) + \frac{2mr^2}{\hbar^2}(E-V)\psi - L^2\psi = 0$

$\psi = R(r)\cdot F(\theta,\phi)$

$$\begin{cases} L^2 F = \ell(\ell+1)F \\ \frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2mr^2}{\hbar^2}(E-V)R = \ell(\ell+1)R \end{cases}$$

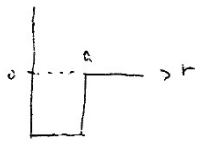
$R = \frac{u}{r}$, $\frac{d^2u}{dr^2} + \left[\frac{2m}{\hbar^2}(E-V) - \frac{\ell(\ell+1)}{r^2}\right]u = 0$

(2) $r<a$ $\qquad k = \sqrt{\frac{2m}{\hbar^2}(E+V_0)}$

$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \left[k^2 - \frac{\ell(\ell+1)}{r^2}\right]R = 0$ or $\frac{d^2u}{dr^2} + \left[k^2 - \frac{\ell(\ell+1)}{r^2}\right]u = 0$

$\underline{\ell=0}$ $\quad \frac{d^2u_0}{dr^2} + k^2 u_0 = 0 \qquad R_0 = \frac{a}{2\pi}\frac{\sin kr}{r}$

$\underline{\ell=1}$ $\quad \frac{d^2R_1}{dr^2} + \frac{2}{r}\frac{dR_1}{dr} + \left(k^2 - \frac{2}{r^2}\right)R_1 = 0 \qquad R_1 = A\frac{dR_0}{dr} = -A\left(\frac{\sin kr}{r^2} - \frac{\cos kr}{r}\right)$



(1) $R_0(a) = 0$

$ka = (n+1)\pi \quad (n = 0,1,2,\cdots)$

$E_{\substack{\ell=0\\ n=0}} = -V_0 + \frac{\hbar^2}{2m}\left(\frac{\pi}{a}\right)^2$

$\exists \;\longleftrightarrow\; \frac{\sqrt{2mV_0}}{\pi\hbar} < a$

$\ell = 1, \quad R_1(a) = 0$

$ka = \gamma_{n-1}\,a + n\pi$

$\forall \ell \qquad R = \frac{w}{\sqrt{r}} \qquad \frac{d^2w}{dr^2} + \frac{1}{r}\frac{dw}{dr} + \left[k^2 - \frac{(\ell+\frac{1}{2})^2}{r^2}\right]w = 0$

$\frac{d^2w}{d(kr)^2} + \frac{1}{kr}\frac{dw}{d(kr)} + \left[1 - \frac{(\ell+\frac{1}{2})^2}{(kr)^2}\right]w = 0 \qquad R_\ell = A_\ell\frac{1}{\sqrt{r}}J_{\ell+\frac{1}{2}}(kr)$ — Bessel fn

$I = \int_0^a |R_\ell|^2 4\pi r^2 dr = 4\pi A_\ell \int_0^a \left(J_{\ell+\frac{1}{2}}(kr)\right)^2 r\,dr = 4\pi A_\ell \frac{a^2}{2}\left\{J'_{\ell+\frac{1}{2}}(ka)\right\}^2$

$\underline{A_1}$ $\quad J'_{\frac{3}{2}}(r) = \sqrt{\frac{2r}{\pi}}\left(\frac{3}{2r}\left(\frac{\sin r}{r^2} - \frac{\cos r}{r}\right) + \frac{\sin r}{r}\right) \quad \ell=1 \quad A_1 = \frac{1}{4a\sin^2 ka} = \frac{a^2+1}{4a^2}$

$J_{\frac{3}{2}}(r) = -\sqrt{\frac{2r}{\pi}}\left(\frac{\sin r}{r^2} - \frac{\cos r}{r}\right)$

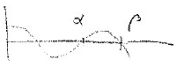
5.4

導体　1s 1p 1d 2s 1f 2p 1g 2d 1h 3s の上

n, ℓ と $n+1, \ell$ の間に存在　② $\ell-1$, ③ $\ell+1$ のうち導体が 1 つ多くある。

☺ $J_\nu(r)$ の r がいくつから zero α, β の間で

$(r^\nu J_\nu)' = r^\nu J_{\nu-1}$ は 1 つ以上 zero をもつ



$(r^{-\nu} J_\nu)' = -r^{-\nu} J_{\nu+1}$ 〃

すこスコ以上 $J_{\nu-1}$ の zero がある r 存在

r 間に J_ν の zero が 1 コあるから 仮定に反する。

$u_{n,\ell} = r R_{n,\ell}$

$$\frac{d^2}{dr^2} u_\ell + \left[\frac{2m}{\hbar^2}(E_\ell - \bar{V}) - \frac{\ell(\ell+1)}{r^2} \right] u_\ell = 0$$

$n_r = 0 \rightarrow E_\ell < E_{\ell+1}$

☹ $W = u_\ell u_{\ell+1}' - u_{\ell+1} u_\ell'$

$W' = u_\ell u_{\ell+1}'' - u_{\ell+1} u_\ell''$

$$= \left[\frac{2m}{\hbar^2}(E_\ell - E_{\ell+1}) + \frac{2\ell+1}{r^2} \right] u_\ell u_{\ell+1}$$

$E_\ell - E_{\ell+1} > 0$ より $W' > 0$

$u_\ell u_{\ell+1} > 0$

$W \rightarrow 0$ ($r=0$, $r=\infty$) より矛盾

$\therefore E_\ell - E_{\ell+1} \leq 0$

6.1

(1) $S=1$, $m_s = 0, \pm 1$ $\quad 2S+1 = 3$ $\quad 3^2 = 9$ (3 y)

(2) $L^2 = (L_1 + L_2)^2$ $\quad L=2$, $M=1$ $\quad L=1$, $M=\begin{cases}0\\-1\end{cases}$ $\quad L = 0, 1, 2$

(3) $L=2$

$|M=2\rangle = |1,1\rangle$

$|M=1\rangle = \frac{1}{2} L_- |L=2, M=2\rangle$

$= \frac{1}{2} (L_{1-} |1,1\rangle + L_{2-} |1,1\rangle)$

$= \frac{1}{2} (\sqrt{2}\,|0,1\rangle + \sqrt{2}\,|1,0\rangle)$

$= \frac{1}{\sqrt{2}} (|0,1\rangle + |1,0\rangle)$

$|J_1, J_2 ; m_1, m_2\rangle$
$= |J_1, m_1\rangle |J_2, m_2\rangle$
$= |1, m_1\rangle |1, m_2\rangle$
$= |m_1, m_2\rangle$

$|M=0\rangle = \frac{1}{\sqrt{6}} L_- |L=2, M=1\rangle = \frac{1}{\sqrt{6}} (|1,-1\rangle + 2|0,0\rangle + |-1,1\rangle)$

$L=1$

$|M=1\rangle = A|1,0\rangle + B|0,1\rangle \perp |L=2, M=1\rangle$ $\quad A^2 + B^2 = 1$, $A>0$

$= \frac{1}{\sqrt{2}} |1,0\rangle - \frac{1}{\sqrt{2}} |0,1\rangle$

$|M=0\rangle = \frac{1}{\sqrt{2}} |1,-1\rangle - \frac{1}{\sqrt{2}} |-1,1\rangle$

$|M=-1\rangle = \frac{1}{\sqrt{2}} |0,-1\rangle - \frac{1}{\sqrt{2}} |-1,0\rangle$

$L=0$

$|M=0\rangle = A|1,-1\rangle + B|0,0\rangle + C|-1,1\rangle$ $\quad A + C + 2B = 0$, $A - C = 0$

$\perp |L=2, M=0\rangle, |L=1, M=0\rangle$

$= \frac{1}{\sqrt{3}} |1,-1\rangle - \frac{1}{\sqrt{3}} |0,0\rangle + \frac{1}{\sqrt{3}} |-1,1\rangle$

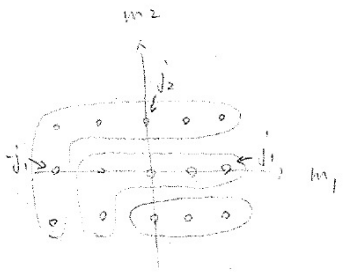
$|L=2, M=-1\rangle = \frac{1}{\sqrt{2}} |-1,0\rangle + \frac{1}{\sqrt{2}} |0,-1\rangle$

$|L=0, M=0\rangle = |-1,-1\rangle$

(4) $(-1)^L$? $\frac{1}{2}$

6.2

(i) $J_- |j_1, j_2; j\, m\rangle = \sqrt{(j+m)(j-m+1)}\, |j_1, j_2; j, m-1\rangle$



$L_1 + L_2 \sim |L_1 - L_2|$

$L^+ |j_1 j_2\, j\, j\rangle = 0$

$|j_1 j_2\, j\, j\rangle = \sum \alpha_{m_1} |m_1\rangle \otimes |j - m_1\rangle$

6.3

(1) 内部スピン $\frac{1}{2}$, $l=2$ $\hat{J}^2$ の固有値 $j(j+1)$ → $j = l+\frac{1}{2}, l-\frac{1}{2} = \frac{5}{2}, \frac{3}{2}$

$j=\frac{5}{2}$ → $m = -\frac{5}{2}, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$

$j=\frac{3}{2}$ → $m = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$

(2) $\hat{H} = \frac{\hat{p}^2}{2m} + V(r) + \xi(\hat{\vec{L}}\cdot\hat{\vec{S}})$

可換 $\begin{cases} \mathbb{L}^2 & L_z \\ \mathbb{S}^2 & S_z \end{cases}$ $\quad \vec{J}^2 = \vec{L}^2 + \vec{S}^2 + 2(\hat{\vec{L}}\cdot\hat{\vec{S}})$ を用いる.

$\boxed{\vec{J}^2\ \vec{L}^2\ \vec{S}^2\ J_z}$

$(\hat{\vec{L}}\cdot\hat{\vec{S}})\phi_0 = \frac{1}{2}\{j(j+1) + l(l+1) - s(s+1)\}\phi_0$

$= \frac{1}{2}\{j(j+1) - \frac{3}{4}\}\phi_0$ $\quad \xi = \text{const}$ と仮定する

$W = \frac{\xi}{2}\{j(j+1) - \frac{3}{4}\}$

$\Delta W = \frac{\xi}{2}\{(j+1)(j+2) - j(j+1)\} = \xi(j+1)$

$\Delta W_{\frac{1}{2}} = \xi\frac{5}{2}$ と書ける

(3) $\psi(\vec{r}) = F(r)\,\Omega_{JM}(\theta,\phi)$

$\Omega_{JM}(\theta,\phi) = \sum\sum Y_{lm}(\theta,\phi)\chi_\mu \langle lsm\mu | ls\,JM\rangle$

$\frac{\hat{p}^2}{2m} + V(r) + \frac{\xi}{2}\{j(j+1) - l(l+1) - \frac{3}{4}\}\psi_{jl}(\vec{r}) = \varepsilon\psi_{jl}(\vec{r})$ $\quad \varepsilon_{nl} = E_{nl} + \frac{\xi}{2}(j(j+1) - \frac{3}{4})$

$U = V + \frac{\xi l}{2}$

$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dF}{dr}\right) - \left\{\frac{l(l+1)}{r^2} + \frac{2m}{\hbar^2}V(r)\right\} = -\frac{2m}{\hbar^2}\left(\varepsilon - \frac{\xi l}{2}\right)F$

7.1

(1) $\psi'(x) = \psi(x') = \hat{U}(\vec{\delta})\psi(x) = \left(1 + \frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{O}}\right)\psi(x)$ ⇒ $\exp\left(\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{P}}\right)$ ← 有限の変換

並進 $\hat{\vec{O}} = \hat{\vec{P}}$ $\psi'(\vec{x}) = \psi(\vec{x}+\vec{\delta}) \simeq (1+\vec{\delta}\cdot\vec{\nabla})\psi(\vec{x}) = \left(1+\frac{i}{\hbar}\vec{\delta}\cdot\vec{p}\right)\psi(\vec{x})$ 微分の変換 ⇒ $\exp\left(\frac{i}{\hbar}\vec{\delta}\cdot(\vec{x}\times\vec{p})\right)$

回転 $\hat{\vec{O}} = \hat{\vec{J}} = \hat{x}\times\hat{p}$ $\psi'(\vec{x}) = \psi(\vec{x}+\vec{\delta}\times\vec{x}) \simeq (1+\vec{\delta}\cdot(\vec{x}\times\vec{\nabla}))\psi(\vec{x}) = \left(1+\frac{i}{\hbar}\vec{\delta}\cdot(\vec{x}\times\vec{p})\right)\psi(\vec{x})$

(2) $\hat{U}(\vec{\delta})\hat{x}_i\hat{U}^\dagger(\vec{\delta}) = \left(1+\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{O}}\right)\hat{x}_i\left(1-\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{O}}^\dagger\right) = \hat{x}_i + \frac{i}{\hbar}\vec{\delta}\cdot\left(\hat{\vec{O}}\cdot\hat{x}_i - \hat{x}_i\hat{\vec{O}}^\dagger\right)$ ($\hat{\vec{O}}^\dagger = \hat{\vec{O}}$)

並進 $= x_i - \frac{i}{\hbar}\vec{\delta}\cdot[x_i, \vec{P}] = x_i + \delta x_i$

回転 $= x_i - \frac{i}{\hbar}\vec{\delta}\cdot[x_i, \vec{J}] = x_i + \delta\omega_{ij}x_j$ $(\vec{p}\times\vec{\delta})_i$

$\hat{U}(\vec{\delta})\hat{P}_i\hat{U}^\dagger(\vec{\delta}) = \left(1+\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{O}}\right)\hat{P}_i\left(1-\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{O}}^\dagger\right) = \hat{P}_i + \frac{i}{\hbar}\vec{\delta}\cdot\left(\hat{\vec{O}}\cdot\hat{P}_i - \hat{P}_i\hat{\vec{O}}^\dagger\right)$ $\hat{\vec{O}}^\dagger = \hat{\vec{O}}$

並進 $= P_i$

回転 $= P_i - \frac{i}{\hbar}\vec{\delta}\,[\vec{P}_i, \hat{\vec{J}}] = P_i - (\vec{p}\times\vec{\delta})_i$

(3) $\hat{U}\hat{H}\hat{U}^\dagger = \hat{H} \iff [\hat{O}, \hat{H}] = 0$

$i\hbar\frac{\partial}{\partial t}\psi'(x) = i\hbar\frac{\partial}{\partial t}\hat{U}(\vec{\delta})\psi(x)$

$= \hat{U}(\vec{\delta})\, i\hbar\frac{\partial}{\partial t}\psi(x)$

$= \hat{U}(\vec{\delta})\hat{H}\psi(x)$

$= \hat{H}\hat{U}(\vec{\delta})\psi(x)$

$= \hat{H}\psi'(x)$

7.2

(1) $\hat{H} = \hat{P}^2/2m$ $\quad \hat{O} = \hat{P}, \hat{L}, \hat{\pi}$ パリティ保存

$\hat{H} = \hat{P}^2/2m + V(\hat{r})$ $\quad \hat{O} = \hat{L}, \hat{\pi}$

$\hat{H} = \hat{P}^2/2m + 1/2m\omega^2(\hat{x}^2+\hat{y}^2) + 1/2m\omega'^2\hat{z}^2$ $\quad \hat{O} = \hat{L}_z, \hat{\pi}$

(2) $[\hat{O}, \hat{H}] = 0 \rightarrow [\hat{U}, \hat{H}] = 0$

(a) $\hat{H}\psi_n' = \hat{H}\hat{U}\psi_n = \hat{U}\hat{H}\psi_n = E_n\psi_n'$

$\hat{H}\psi_n' = \hat{H}\left(1 + \frac{i}{\hbar}\hat{\vec{O}}\cdot\vec{\delta}\right)\psi_n(x)$

$= \left(1 + \frac{i}{\hbar}\hat{\vec{O}}\cdot\vec{\delta}\right)\hat{H}\psi_n(x)$

$= E_n\psi_n'$

(b) (i) 平面波

(ii) 中心力場の解

(iii) 球関数

7.3

(1) $H = \frac{1}{2m}(P_x^2 + P_y^2 + P_z^2) + \frac{1}{2}m\omega^2(x^2+y^2+z^2)$

$n_x = 1,\ n_y = n_z = 0 \qquad \Psi(x,y,z) = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \hat{a}_x^\dagger e^{-\frac{m\omega}{2\hbar}(x^2+y^2+z^2)}$

$\hat{U}(\theta) = e^{-i\theta\hat{L}_z} \qquad \hat{L}_z = \frac{1}{i}(\hat{a}_x^\dagger \hat{a}_y - \hat{a}_x \hat{a}_y^\dagger)$

$\hat{L}_z^n |100\rangle = \begin{cases} i|010\rangle & (n\text{ odd}) \\ |100\rangle & (n\text{ even}) \end{cases}$

$\hat{L}_z^n |010\rangle = \begin{cases} -i|100\rangle & (n\text{ odd}) \\ |010\rangle & (n\text{ even}) \end{cases}$

$\hat{L}_z^n |001\rangle = \begin{cases} |001\rangle & n=0 \\ 0 & n>0 \end{cases}$

$\hat{U}(\theta)|100\rangle = \cos\theta|100\rangle + i\sin\theta|010\rangle$

$\hat{U}(\theta)|010\rangle = -i\sin\theta|100\rangle + \cos\theta|010\rangle$

$\hat{U}(\theta)|001\rangle = |001\rangle$

(2) $R(\pi)\ Y_{lm}(\theta,\varphi)$

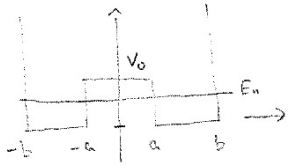
$\Downarrow$

$R(\pi)\ Y_{lm}(\pi-\theta, \varphi+\pi)$

$(-1)^l Y_{lm}(\theta,\varphi)$

$\hat{U}_{\text{ﾈﾝﾃﾝｾﾝ}} \propto \hat{L}$ は時間反転で符号が変わる

7.4



$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

(1) $E_n \ll V_0$

$|S\rangle$



$|A\rangle$



$E_S < E_A$

7.4

$k \gg K \quad ka \sim n \quad ka \gg n$

対称解



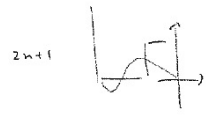
$\psi = \begin{cases} A\cos K(x+b) \\ S\sin kx \end{cases}$

$A^2 - \left(\frac{K}{k}\right)^2 \cos^2 = S^2 \qquad S = \frac{\frac{K}{k}\cos}{\sqrt{\sin^2 - 1}} \qquad \therefore A\cos k(b-a) \sim \frac{k}{K}\sin k(b-a)$

$\wr$

$(k_s - k_a)a$

反対称



$\psi = \begin{cases} A\cos K(x+b) \\ A\sin kx \end{cases}$

$A^2 - \left(\frac{K}{k}\right)^2 \cos^2 = A^2 \qquad A = \frac{\frac{K}{k}\cos}{\sqrt{\sin^2 + 1}} \qquad \therefore A\cos k(b-a) \sim \frac{k}{K}\cos K(b-a)$

$k_s(b-a) = 2\pi n$

(2) $\psi_0 = 2A\cos k_0(x+b)$

$\psi_s - \psi_a \sim \psi_0(-x)$

$\psi_s(x) + \psi_a(x) \sim \psi_0(x)$

$e^{-i\omega_s t}\psi_s + e^{-i\omega_a t}\psi_a \sim \psi_0(x,t)$

$=$

$\left(e^{-i\omega_s t} + e^{-i\omega_a t}\right)\psi_0(x) + \left(e^{-i\omega_s t} - e^{-i\omega_a t}\right)\psi_0(-x)$

$=$

$2e^{-i\Omega t}\left(\cos\Delta\omega t\ \psi_0(x) - A\sin\Delta\omega t\ \psi_0(-x)\right)$

$\Delta\omega = \omega_s - \omega_a$

$\left(\omega - \frac{I}{\hbar}\right)$

7.5

(1) $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V(x)\psi(x) = E\psi(x)$

$$\begin{pmatrix} u_1(x+L) & u_1'(x+L) \\ u_2(x+L) & u_2'(x+L) \end{pmatrix} = A \begin{pmatrix} u_1(x) & u_1'(x) \\ u_2(x) & u_2'(x) \end{pmatrix}$$

= ホン determinant が ロンスケアン

$|u_1\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ $|u_2\rangle = \begin{pmatrix}0\\1\end{pmatrix}$

$x' = x + L$ ノ並進ニ連結子 $\hat{U}_L$

$[H, U_L]$ $U_L|u_i\rangle \in V$

$\hat{U}_L|u_1\rangle = \begin{pmatrix}a_{11}\\a_{21}\end{pmatrix}$ $U_L|u_2\rangle = \begin{pmatrix}a_{12}\\a_{22}\end{pmatrix}$ $U_L = \begin{pmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{pmatrix} = A$ (unitaly)

固有ケ $|u\rangle$ ($\hat{U}_L$ ノ eigen ket)

$U_L|u\rangle = e^{i\theta}|u\rangle$

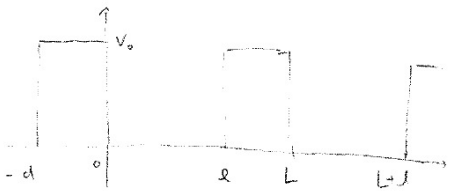
$\lambda^2 - (a_{11}+a_{22})\lambda + 1 = 0$

$\langle x|U_L|u\rangle = e^{i\theta}\langle x|u\rangle$

$\therefore\ u(x+L) = e^{i\theta}u(x)$

$u(x) = e^{ikx}v(x)$ $(k = \theta/L)$ $\longrightarrow$ $v(x+L) = v(x)$ //

(2)



$$\begin{vmatrix} 1 & 1 & 1 & -1 \\ k & -k & -ik & ik \\ e^{ikL-kd} & e^{ikL+kd} & -e^{ik\ell} & -e^{-ik\ell} \\ ke^{ikL-kd} & -ke^{ikL+kd} & -ike^{-ik\ell} & ike^{-ik\ell} \end{vmatrix} = 0$$

$f(E) = \cosh kd \cos k\ell + \frac{K^2-k^2}{2kd}\sinh kd \sin k\ell = \cos kL$ $\qquad K = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

$k = \frac{\sqrt{2mE}}{\hbar}$

(i) $E < V_0$

$k\ell = n\pi$ $\qquad E = \frac{\hbar^2}{2m}\left(\frac{n\pi}{\ell}\right)^2$

$f(E) = (-1)^n \cosh kd$ $\qquad kd > 0$ ナリ $\cosh kd > 1$ $\longrightarrow$ $|f(E)| > 1$ $\qquad \therefore E \neq \frac{\hbar^2}{2m}\left(\frac{n\pi}{2m}\right)^2$

(ii) $E \gg V_0$

$i\sigma = K$ $\longrightarrow$ $f(E) = \cos\sigma d\cos k\ell - \frac{\sigma^2+k^2}{2\sigma k}\sin\sigma d\sin k\ell$

$\sigma d + k\ell = 2n\pi$ のとき $f(E) > 1$

$= (2n+1)\pi$ 〃 < -1

8.1 $H = H_0 + \epsilon H_1$

$E_n = E_n^{(0)} + \epsilon E_n^{(1)} + \epsilon^2 E_n^{(2)} + \cdots$

$E_n^{(1)} = \langle n|H_1|n\rangle$

$E_n^{(2)} = \sum_{m\neq n} \frac{|\langle n|H_1|m\rangle|^2}{E_n^0 - E_m^0}$

1) $\alpha = \sqrt{\frac{m\omega}{\hbar}}$ $\quad \alpha x|n\rangle = \sqrt{\frac{n}{2}}|n-1\rangle + \sqrt{\frac{n+1}{2}}|n+1\rangle$

$\alpha^2 x^2|n\rangle = \frac{\sqrt{n(n-1)}}{2}|n-2\rangle + \frac{2n+1}{2}|n\rangle + \frac{\sqrt{(n+1)(n+2)}}{2}|n+2\rangle$

$E_n^{(1)} = \alpha\langle n|x^2|n\rangle = \frac{\alpha}{\alpha^2}\frac{2n+1}{2} = \alpha\left(n+\frac{1}{2}\right)\frac{\hbar}{m\omega}$ //

2) $E_n^{(1)} = eE\langle n|x|n\rangle = 0$ 2次の摂動

$E_n^{(2)} = e^2E^2 \sum \frac{|\langle n|x|m\rangle|^2}{E_n - E_m} = -\frac{e^2E^2}{2m\omega^2}$

$\langle n|x|m\rangle = \frac{1}{\alpha}\left(\sqrt{\frac{n}{2}}\delta_{n,m+1} + \sqrt{\frac{n+1}{2}}\delta_{n,m+1}\right)$

3) $\alpha^4\langle n|x^4|n\rangle = \frac{1}{4}(6n^2+6n+3)$

$E_n^{(1)} = \beta\langle n|x^4|n\rangle = \beta\frac{\hbar^2}{m^2\omega^2}\cdot\frac{6n^2+6n+3}{4}$

最密解

1) $H = -\frac{\hbar^2}{2m}\nabla^2 + \underbrace{\left(\frac{1}{2}m\omega^2 + \alpha\right)}_{=\frac{1}{2}m\omega'^2}x^2$ $\quad \omega' = \omega\sqrt{1+\frac{2\alpha}{m\omega^2}} \simeq \omega + \frac{\alpha}{m\omega}$

$\Delta E_n = \left(n+\frac{1}{2}\right)\hbar\Delta\omega = \left(n+\frac{1}{2}\right)\frac{\alpha\hbar}{m\omega}$

2) $H = -\frac{\hbar}{2m}\nabla^2 + \frac{1}{2}m\omega^2x^2 + eEx$

$= -\frac{\hbar^2}{2m}\nabla^2 + \frac{1}{2}m\omega\left(x + \frac{eE}{m\omega^2}\right)^2 - \frac{e^2E^2}{2m\omega^2}$

8.2

(1) $\int \Psi_0 e\vec{E}\cdot\vec{r}\,\Psi_0\,d\tau = 0$ (対称性) $\quad P\cdot E = 0$

(2) $E_0^{(2)} = \sum_{n\neq 0} \frac{|\langle 0|\hat{H}_1|n\rangle|^2}{E_0^{(0)} - E_n^{(0)}} \simeq \sum_{n\neq 0} \frac{\langle 0|H_1|n\rangle\langle n|H_1|0\rangle}{E_0^{(0)} - E_n^{(0)}} = \sum_{n=0}^{\infty} \frac{\langle 0|H_1|n\rangle\langle n|H_1|0\rangle - \langle 0|H_1''|0\rangle\langle 0|H_1|0\rangle}{E_0^{(0)}}$

(次の摂動)

$$= \frac{\langle 0|H_1^2|0\rangle}{E_0^{(0)}} = \frac{1}{E_0^{(0)}}\frac{e^2E^2}{a_0^2\pi^2}$$

(3) $\Psi = \sum_{n\beta} a_{n\beta}\phi_{n\beta}$

$H = H_0 + \epsilon H_1$

$\langle m\alpha|$ をかける $\quad a_{m\alpha}E_m^0 + \epsilon a_{n\beta}\langle m\alpha|H_1|n\beta\rangle = a_{m\alpha}E$

0 でない $a_{k\gamma} = 0 \quad k\neq m$

$\therefore n = m$ のみ

$$a_{m\alpha}(E_m^0 - E) + \epsilon a_{n\beta}\langle m\alpha|H_1|m\beta\rangle = 0$$ ← 永年方程式

$a_{m\alpha}\neq 0 \rightarrow$ determinant $= 0$

H-atom $n=2$

$E \parallel e_z$ となる $x'y'z'$ 系

$\epsilon H_1 = -eEz$

$$\Psi_1 = \Psi_{200} = \frac{1-\frac{r}{2a_0}}{\sqrt{2a_0^3}} e^{-\frac{r}{2a_0}} Y_{00}$$

$$\Psi_{2,3,4} = \Psi_{21,0,-1} = \frac{\frac{r}{2a_0}}{\sqrt{6a_0^3}} e^{-\frac{r}{2a_0}} Y_{1,0,-1}$$

$\langle 1|H_1|2\rangle = \langle 2|H_1|1\rangle = 3eEa_0 = V$

$$\begin{vmatrix} E_2^0 - E & V & & \\ V & E_2^0 - E & & \\ & & E_2^0 - E & \\ & & & E_2^0 - E \end{vmatrix} = 0$$

$E = E_2^0, E_2^0, E_2^0 \pm V$

$E_a \quad E_b \quad E_{c,d}$

$\Psi_{ab} = \alpha\Psi_{211} + \beta\Psi_{21-1} \qquad (\alpha^2+\beta^2=1)$

$\Psi_c = \frac{1}{\sqrt{2}}(\Psi_{200} + \Psi_{210})$

$\Psi_d = \frac{1}{\sqrt{2}}(\Psi_{200} - \Psi_{210})$

8.3

(1) $\psi(\phi,t) = \psi e^{-iEt/\hbar}$ $\quad \frac{\partial}{\partial\phi}\psi + \frac{2IE}{\hbar^2}\psi = 0$ $\quad \psi = \frac{1}{\sqrt{2\pi}} e^{i\frac{\sqrt{2IE}}{\hbar}\phi}$

$E = \frac{n^2\hbar^2}{2I}$ $(n = 0, \pm1, \pm2, \cdots)$ $\quad \psi = \frac{1}{\sqrt{2\pi}} e^{in\phi}$

(2) $\frac{1}{\sqrt{2\pi}} e^{in\phi} = |n\rangle$

$|\psi_n, s\rangle = c_1|n\rangle + c_2|-n\rangle + \lambda|\psi_n^{(1)} s\rangle + \lambda^2|\psi_n^{(2)} s\rangle + \cdots$

$(s = n, -n)$

$|\psi_n^{(0)}, s\rangle = c_1^{(0)}|n\rangle + c_2^{(0)}|-n\rangle + \sum_{\ell\neq n} c'^{(0)}_{n,\ell}|\phi_\ell\rangle$

$|\psi_n^{(1)} s\rangle =$

$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} \cdots$

$\hat{V} = -\mu E\cos\phi$

$(\hat{H}_0 + \lambda\hat{\bar{V}})|\psi_n, s\rangle = E_n|\psi_n, s\rangle$

$$\begin{cases} c'^{(1)}_{n,\ell} = \dfrac{\hat{\bar{V}}_{\ell,n} c_1 + \bar{V}_{\ell,-n} c_2}{E_n^{(0)} - E_\ell^{(0)}} \\ c_1(\bar{V}_{nn} - E_n^{(1)}) + c_2 V_{n,-n} = 0 \\ c_1 V_{-n,n} + c_2(V_{n,-n} - E_n^{(1)}) = 0 \quad \to \quad E_n^{(1)} = 0 \end{cases}$$

但し $V_{\pm n,\pm n} = \langle \pm n|\bar{V}|\pm n\rangle$

$$\bar{V}_{n\ell}\sum_{\ell\neq n} C_{n\ell}^{(1)} - c_1 E_n^{(2)} = 0$$

$$\bar{V}_{-n\ell}\sum_{\ell\neq n} C_{n\ell}^{(1)} - c_2 E_n^{(2)} = 0$$

$$\begin{vmatrix} \sum \dfrac{V_{n\ell}V_{\ell n}}{E_n^{(0)} - E_\ell^{(0)}} - E_n^{(2)} & \sum \dfrac{V_{n\ell}V_{\ell,-n}}{} \\ \sum \dfrac{V_{-n\ell}V_{\ell n}}{} & \sum \dfrac{V_{-n\ell}V_{\ell n}}{} - E_n^{(2)} \end{vmatrix} = 0$$

$$\sum_{\ell\neq n} \frac{V_{n\ell}V_{\ell n}}{E_n^{(0)} - E_\ell^{(0)}} = \frac{I\mu^2E^2}{\hbar^2(4n^2-1)} \qquad \sum_{\ell\neq n} \frac{V_{n\ell}V_{\ell,-n}}{E_n^{(0)} - E_\ell^{(0)}} = \begin{cases} \dfrac{I\mu E^2}{2\hbar^2} & n = \pm1 \\ 0 & n \neq \pm1 \end{cases}$$

(i) $n=0$ $\quad E_0^{(2)} = \frac{I\mu^2E^2}{\hbar^2}$

(ii) $n=\pm1$ $\quad E_{\pm1}^{(2)} = \frac{5}{6}\frac{I\mu E^2}{\hbar^2}, -\frac{1}{6}\frac{I\mu E^2}{\hbar^2}$

(iii) the other $\quad E_n^{(2)} = \frac{I\mu E^2}{\hbar^2(4n^2-1)}$

固有関数

$c_1 = c_2 = \frac{1}{\sqrt{2\pi}}\cos\phi$

$c_1 = -c_2 = \frac{1}{\sqrt{2\pi}}\sin\phi$

8.4

$$\hat{H} = \hat{H}_0 + \hat{H}_1$$

$$H_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2$$

$$H_1 = \cdot\, d(t)\,\hat{x}^2 \qquad d(t) = \frac{m\omega^2}{2}\epsilon\cos\Omega t$$

(1) $\hat{H}_0\psi_n = E_n\psi_n$

$$\psi(t) = \sum c_n(t)\psi_n e^{-iE_nt/\hbar}$$

$$\hat{H}\psi(t) = i\hbar\frac{\partial\psi}{\partial t}$$

$$\rightarrow \sum \underline{c_n(t)}\left(E_n + \cdot\, d(t)\,\hat{x}^2\right)\psi_n e^{-iE_nt/\hbar} = i\hbar\sum\left(c_n'(t) - i\,c_n(t)E_n/\hbar\right)\psi_n e^{-iE_nt/\hbar}$$

$$= \sum\left(i\hbar\,c_n'(t) + \underline{c_n(t)E_n}\right)\psi_n e^{-iE_nt/\hbar}$$

$$\frac{m\omega^2}{2}\epsilon\sum c_n(t)\,d(t)\,\hat{x}^2\cdot\psi_n e^{-iE_nt/\hbar} = i\hbar\sum c_n'(t)\psi_n e^{-iE_nt/\hbar}$$

$\int dx\,\psi_m^* e^{iE_mt/\hbar}$ をかけると $\quad \langle m|\hat{x}^2|n\rangle = \frac{\hbar}{2m\omega}\left(\sqrt{n(n-1)}\,\delta_{n,m-2} + (2n+1)\delta_{nm} + \sqrt{(n+1)(n+2)}\,\delta_{n,m+2}\right)$

$$c_m'(t) = -\frac{i\,d(t)}{2m\omega}\left(\sqrt{n(n-1)}\,c_{n-2}(t)\,e^{i(E_m-E_{n-2})t/\hbar} + \sqrt{(n+1)(n+2)}\,c_{n+2}(t)\,e^{i(E_m-E_{n+2})t/\hbar}\right)$$

$$= -\frac{i\omega\epsilon}{4}\cos\Omega t\left[\sqrt{n(n-1)}\,c_{n-2}(t)\,e^{2i\omega t} + (2n+1)c_n(t) + \sqrt{(n+1)(n+2)}\,c_{n+2}(t)\,e^{-2i\omega t}\right]$$

(2) ϵ について2

0次 $\dot{c}_n^{(0)}(t) = 0$ $\quad$ -- ①

1次 $\dot{c}_n^{(1)}(t) = -\frac{i\omega}{4}\cos\Omega t\left[\sqrt{n(n-1)}\,c_{n-2}^{(0)}(t)\,e^{2i\omega t} + (2n+1)c_n^{(0)}(t) + \sqrt{(n+1)(n+2)}\,c_{n+2}^{(0)}(t)\,e^{-2i\omega t}\right]$

-- ②

(3)

8.4

(3) 摂動系列 $C_0^{(0)}(0)=1$ $C_n^{(0)}(0)=0$ $(n\geq 1)$ $C_n^{(s)}(0)=0$ $(n\geq 0,\ s\geq 1)$ より

① 式1) $C_0^{(0)}(t)=1$, $C_n^{(0)}=0$ $(n\geq 1)$ となる

② $C_0^{(1)}(t)=-\frac{i\omega}{4}\cos\Omega t$ $\qquad C_0^{(1)}(t)=-\frac{i\omega}{4\Omega}\sin\Omega t$ $(\Omega\neq 0)$

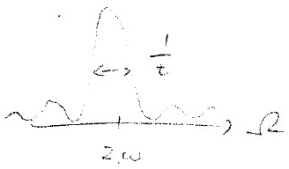
$$\dot{C}_2^{(1)}(t)=-\frac{i\omega}{4}\cos\Omega t\,\sqrt{2}\,e^{2i\omega t}=-\frac{i\omega}{4\sqrt{2}}\left[e^{i(2\omega+\Omega)t}+e^{i(2\omega-\Omega)t}\right]$$

$\Omega\neq\pm 2\omega$ のとき $\quad C_2^{(1)}(t)=-\frac{\omega}{4\sqrt{2}}\left[\frac{1}{2\omega+\Omega}e^{i(2\omega+\Omega)t}+\frac{1}{2\omega-\Omega}e^{i(2\omega-\Omega)t}\right]+\frac{\omega}{4\sqrt{2}(4\omega^2-\Omega^2)}$

$\Omega=\pm 2\omega$ $\quad C_2^{(1)}(t)=-\frac{i\omega}{4\sqrt{2}}t-\frac{1}{16\sqrt{2}}e^{\pm 2i\omega t}\cdot\frac{1}{16\sqrt{2}}$

$C_n^{(1)}(t)=0$ $(n\neq 0,2)$

$$|c|^2\quad c\left(\frac{e^{i(2\omega+\Omega)t}-1}{2\omega+\Omega}-\frac{e^{i(2\omega-\Omega)t}}{2\omega-\Omega}\right)\sim\frac{\Omega^2\frac{1}{2}(2\omega-\Omega)t}{(2\omega)^2-\Omega^2}$$



9.1

(1) $m_1 = \sum_n (E_n - E_0) |\langle n|\hat{Q}|0\rangle|^2 = \sum (E_n - E_0) \langle 0|\hat{Q}^\dagger|n\rangle\langle n|\hat{Q}|0\rangle$

$= \sum \langle 0|\hat{Q}\hat{H}|n\rangle\langle n|\hat{Q}|0\rangle - \frac{1}{2}\{\langle 0|\hat{H}\hat{Q}^2|0\rangle + \langle 0|\hat{Q}^2\hat{H}|0\rangle\}$

$= \langle 0|\hat{Q}\hat{H}\hat{Q}|0\rangle - \frac{1}{2}\langle 0|(\hat{H}\hat{Q}^2 + \hat{Q}^2\hat{H})|0\rangle$ //

$\left(\because [Q,[\hat{H},\hat{Q}]] = 2\hat{Q}\hat{H}\hat{Q} - \hat{H}\hat{Q}^2 - \hat{Q}^2\hat{H}\right)$

(2) $[\hat{x},[\hat{H},\hat{x}]] = \frac{1}{2m}[\hat{x},[\hat{p}^2,\hat{x}]] = \frac{\hbar^2}{m}$ $\qquad m_1 = \frac{\hbar^2}{2m}\langle 0|0\rangle$

(3) $[\hat{x}^2,[\hat{H},\hat{x}^2]] = \frac{1}{2m}[\hat{x}^2,[\hat{p}^2,\hat{x}^2]] = \frac{2i\hbar}{m}[\hat{p}\hat{x},\hat{x}^2] = \frac{4\hbar^2}{m}\hat{x}^2$

$\quad = -2i\hbar(\hat{x}\hat{p}+\hat{p}\hat{x}) \qquad = -2i\hbar\hat{x}^2$

$\quad = -4i\hbar\hat{p}\hat{x} + 2\hbar^2$

$m_1 = \frac{4\hbar^2}{m}\langle 0|\hat{x}^2|0\rangle$ //

(4) $m_3 = \sum (E_n^3 - 3E_n^2E_0 + 3E_nE_0^2 - E_0^3)\langle 0|Q|n\rangle\langle n|Q|0\rangle$

$= \sum$

10.1

(1) $i\hbar \frac{\partial}{\partial t}\left(\hat{U}(t)\,\Psi_\alpha^S(x,t=0)\right) = \hat{H}\left(\hat{U}(t)\,\Psi_\alpha^S(x,t=0)\right)$

$\left(i\hbar \frac{\partial \hat{U}(t)}{\partial t}\right)\Psi_\alpha^S(x,t=0) = \hat{H}\hat{U}(t)\Psi_\alpha^S(x,t=0)$

$i\hbar \frac{\partial \hat{U}}{\partial t} = \hat{H}\hat{U} \qquad \hat{U} = e^{-i\hat{H}t/\hbar}$

(2) $\frac{d}{dt}\langle \Psi_\beta^S(x,t)|\hat{\Omega}_S|\Psi_\alpha^S(x,t)\rangle$

$i\hbar \frac{\partial}{\partial t}|\Psi_\alpha^S(x,t)\rangle = \hat{H}|\Psi_\alpha^S(x,t)\rangle$ より $\left(\frac{iE_\beta}{\hbar} - \frac{iE_\alpha}{\hbar}\right)\langle \Psi_\beta^S(x,t)|\hat{\Omega}_S|\Psi_\alpha^S(x,t)\rangle$

(3) $\frac{d}{dt}\langle \Psi_\beta^S(x,t)|\Omega_S|\Psi_\alpha^S(x,t)\rangle$

$= \langle \Psi_\beta^S(x,t=0)|\frac{d}{dt}\left(e^{iHt/\hbar}\,\Omega_S\,e^{-iHt/\hbar}\right)|\Psi_\alpha^S(x,t=0)\rangle$

$= \langle \Psi_\beta^S(x,t=0)|\frac{i}{\hbar}\hat{H}e^{i\hat{H}t/\hbar}\Omega_S e^{-i\hat{H}t/\hbar} + e^{iHt/\hbar}\Omega_S e^{-iHt/\hbar}\frac{-i}{\hbar}\hat{H}|\Psi_\alpha^S(x,t=0)\rangle$

$= \langle \Psi_\beta^S|\frac{1}{i\hbar}[\Omega_H, H]|\Psi_\alpha^S\rangle$

$\Psi_\alpha^S(x,t=0)$ $\Psi_\beta^S(x,t=0)$ は任意にとれるので $\frac{d}{dt}\Omega_H = \frac{1}{i\hbar}[\Omega_H, H]$

(4)

10.2

(1) $\hat{H} = -\hat{\vec{\mu}} \cdot \vec{H} = -H\hat{\mu}_z = -\gamma\hbar H \hat{J}_z$ $\qquad \vec{H} = (0, 0, H)$

(2) $[\hat{H}, \hat{J}] = -\gamma\hbar H [\hat{J}_z, \hat{J}]$

$$\left.\begin{aligned} [\hat{H}, \hat{J}_z] &= -i\gamma\hbar H J_y \\ [\hat{H}, \hat{J}_y] &= -i\gamma\hbar H J_x \\ [\hat{H}, \hat{J}_x] &= 0 \end{aligned}\right\} \rightarrow \left\{\begin{aligned} \frac{d\hat{J}_{xH}(t)}{dt} &= \gamma H \hat{J}_{yH}(t) \\ \frac{d\hat{J}_{yH}(t)}{dt} &= -\gamma H \hat{J}_{xH}(t) \\ \frac{d\hat{J}_{zH}(t)}{dt} &= 0 \end{aligned}\right\} \quad \left\{\begin{aligned} \hat{J}_{xH}(0) &= J_x \\ \hat{J}_{yH}(0) &= J_y \\ \hat{J}_{zH}(0) &= J_z \end{aligned}\right\}$$

$J_{xH}(t) = J_x \cos(\gamma H t) + J_y \sin(\gamma H t)$

$J_{yH}(t) = J_y \cos(\gamma H t) - J_x \sin(\gamma H t)$

$J_{zH}(t) = J_z$

(3) $j = \frac{1}{2}$ $\qquad |+\rangle = |j = \frac{1}{2}, j_z = \frac{1}{2}\rangle$, $|-\rangle = |j = \frac{1}{2}, j_z = -\frac{1}{2}\rangle$ $\}$ (ｽﾋﾟﾝ ｱｯﾌﾟ ﾀﾞｳﾝ) は完全直交系

$\langle + | + \rangle = \langle - | - \rangle = 1$

$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\qquad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$J_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\qquad J_{xH}(t) = \frac{1}{2}\begin{pmatrix} 0 & e^{-i\gamma H t} \\ e^{i\gamma H t} & 0 \end{pmatrix}$

$J_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\qquad J_{yH}(t) = \frac{1}{2}\begin{pmatrix} 0 & -e^{-i\gamma H t} \\ e^{i\gamma H t} & 0 \end{pmatrix}$

$J_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ $\qquad J_{zH} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

10.3

(1) $\psi' = e^{i\frac{e}{\hbar c}\chi}\psi \qquad (\vec{p} - \frac{e}{c}\vec{A}')\psi' = (\vec{p} - \frac{e}{c}\vec{A})\psi$

$H'\psi' = e^{i\frac{e}{\hbar c}\chi} E\psi = E\psi'$

(2) $\langle x \rangle = \int \psi'^* x \psi' d\tau = \int \psi^* e^{-i\frac{e}{\hbar c}\chi} x e^{i\frac{e}{\hbar c}\chi} \psi d\tau = \int \psi^* x \psi d\tau$

p 는 불변량

$p' = p - \frac{e}{c}A$

(3) $\vec{A} = (-Hy, 0, 0)$

$\hat{H} = \frac{1}{2m}\left\{ -\hbar^2\nabla^2 - 2i\hbar\frac{e}{c}By\frac{\partial}{\partial x} + \frac{e^2}{c^2}B^2y^2 \right\}$

$\psi = e^{i(\alpha x + \beta z)}\varphi(z)$

$\left\{ -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2} + \frac{\alpha\hbar eB}{mc}y + \frac{e^2B^2}{2mc^2}y^2 \right\}\varphi(y) = \left\{ E - \frac{\hbar^2}{2m}(\alpha^2+\beta^2) \right\}\psi$

$\frac{e^2B^2}{2mc^2}\left(y + \frac{c\alpha\hbar}{eB}\right)^2 - \frac{\alpha^2\hbar^2}{2m} \qquad y' = y + \frac{c\alpha\hbar}{eB}$

$\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2} + \frac{e^2B^2}{2mc^2}y'^2 \right)\varphi(y) = \left(E - \frac{\hbar^2\beta^2}{2m}\right)\varphi(y)$

$h = 0 \qquad E - \frac{\hbar^2\beta^2}{2m} = \left(\frac{1}{2} + n\right)\hbar\omega$

$\omega^2 = \frac{e^2B^2}{mc^2}$